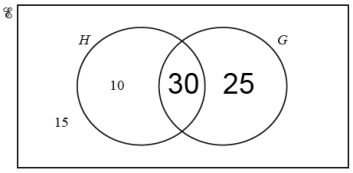
Question	Answer	Marks	Partial Marks
14(c)	$\frac{12+13}{2}$ oe leading to 12.5 nfww	**1**	
15	100	**3**	**M2** for 180 – (180 –140) × 2 oe or **M1** for 180 – 140 or for 180 – (2 × *their* 40)
16(a)	–4.5 3	**2**	**B1** for each
16(b)	Correct graph drawn	**4**	**B3FT** for 9 points correctly plotted or **B2FT** for 7 points correctly plotted or **B1FT** for 5 correctly plotted
17(a)	23 or 29	**1**	
17(b)	9 cao	**1**	
18	$\frac{7}{18}$ cao	**2**	**M1** for $\frac{11}{18} - \frac{4}{18}$ oe
19	$\frac{5\times20}{2}$	**M1**	
	50 cao	**A1**	If 0 scored **SC1** for 2 correct roundings or for all 3 correct but with any trailing zeros
20	125	**2**	**M1** for 75 ÷ 3 [× 5] oe
21(a)	Rotation 90 anticlockwise oe (0,0) oe	**3**	**B1** for rotation **B2** for 90 anticlockwise **and** (0, 0) or for 90 clockwise **and** (0.5, 3.5) **or B1** for 90 anticlockwise **or** 90 clockwise **or** (0, 0) **or** (0.5, 3.5)
21(b)	Rectangle with vertices at (1, –3) (3, –3) (3, –4) (1,–4)	**2**	**B1** for reflection in $y = k$ or in $x = -1$
22(a)	200	**1**	

Question	Answer	Marks	Partial Marks
22(b)	No and a correct statement based on correct reasoning or calculations	**2**	**B1** for a correct statement without support e.g. 30g is needed for 18 cookies oe or 2g more [than 28g] for 18 cookies oe or 28g would only give 16 cookies oe or 8g would only give 4 more cookies oe or **M1** for correct calculation $\frac{18}{12}\times 20$ oe or $\frac{6}{12}\times 20$ oe or $\frac{28}{20}\times 12$ oe or $\frac{8}{20}\times 12$ oe or $\frac{20}{12}$ **and** $\frac{28}{18}$ oe
23(a)	7 cao	**1**	
23(b)	3 cao	**1**	
24(a)		**2**	**B1** for one correct value correctly placed If 0 scored, **SC1** for two positive integers such that n(G) = 55
24(b)	55	**1**	**FT** *their* 30 + *their* 25 provided *their* 30 + *their* 25 is less than 80
24(c)	$\frac{15}{80}$ oe	**1**	
25	$x^2 + x - 6$ final answer	**2**	**B1** for $x^2 + 3x - 2x - 6$ oe with at least three terms correct
26	175 300 000 cao	**1**	
27	$8\pi + 16$ or $8(\pi + 2)$ final answer	**3**	**M2** for [perimeter =] $\frac{8+8}{2}\times\pi + 8 + 8$ oe or $8\pi + 16$ oe seen and then spoilt or **M1** for [arc =] $\frac{8+8}{2}\times\pi$ oe